SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – SHORT ANSWER TEST (5 QUESTIONS)

Course Code:	ITA0509	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 1: Short Answer Test
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
Student Name:	Vimal. R	**Reg. No.:**	192421350
Section / Batch:	2024.	**Date:**	05. 10. 2026

Code of Conduct

I Vimal. R 192421350 *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student:

Vimal. R

QUESTIONS: 5

Total Marks:50

Annexure A

Short Answer Test

Q1. Smart Traffic Monitoring System

A smart city is planning to install cameras at major traffic junctions to monitor vehicles, pedestrians, and traffic conditions. The captured images may vary in resolution, lighting, and camera position. **Apply the fundamental concepts of Computer Vision to design a suitable processing pipeline for this traffic monitoring system. Explain the major stages from image acquisition to object detection and classification.**

Q2. Automated Attendance System

A college wants to implement an automated attendance system using cameras that capture students sitting inside classrooms. The images may contain different lighting conditions, face orientations, shadows, and multiple students.

Analyze how image representation, preprocessing, feature extraction, and face recognition can be applied to develop the proposed attendance system. Discuss the major challenges that may affect its accuracy.

Q3. Medical Image Analysis System

A hospital plans to use a Computer Vision system to assist doctors in analyzing X-ray images for identifying abnormal regions. The images may contain noise, poor contrast, and different brightness levels.

Evaluate the importance of image resolution, intensity, preprocessing, segmentation, and feature extraction in this application. Explain how these factors can influence the accuracy and reliability of the system.

Q4. Security Surveillance System

A shopping mall wants to develop an intelligent surveillance system using cameras installed at entrances, corridors, and parking areas. The system must identify people and objects under different lighting and crowded conditions.

Analyze how the fundamental stages of a Computer Vision system can be used for this application.

Explain the effects of illumination, occlusion, camera position, and image quality on the final results.

Q5. Smart Agriculture Monitoring

A farming organization wants to use drone cameras to identify healthy plants, damaged plants, weeds, and disease-affected areas in large agricultural fields. The captured images may contain shadows, different lighting conditions, and variations in plant size.

Q1: Smart traffic Monitoring system.

A computer vision - based traffic monitoring system can be designed using the following stages.

Image Acquisition :- cameras installed at major traffic Junctions continously capture img or ved of the vehicles

Preprocessing :- Images are resized, noise is removed, and brightness and contrast are adjust to handle different lighting conditions.

Segmentation :- The road scene is separated into foreground objects and background to identify vehicles and pedestrians.

Feature extraction :- Features such as shape, colour, edges, texture and size are extracted from detected objects.

Object Detection :- computer vision algorithms detect and locate vehicles and pedestrians in the captured images.

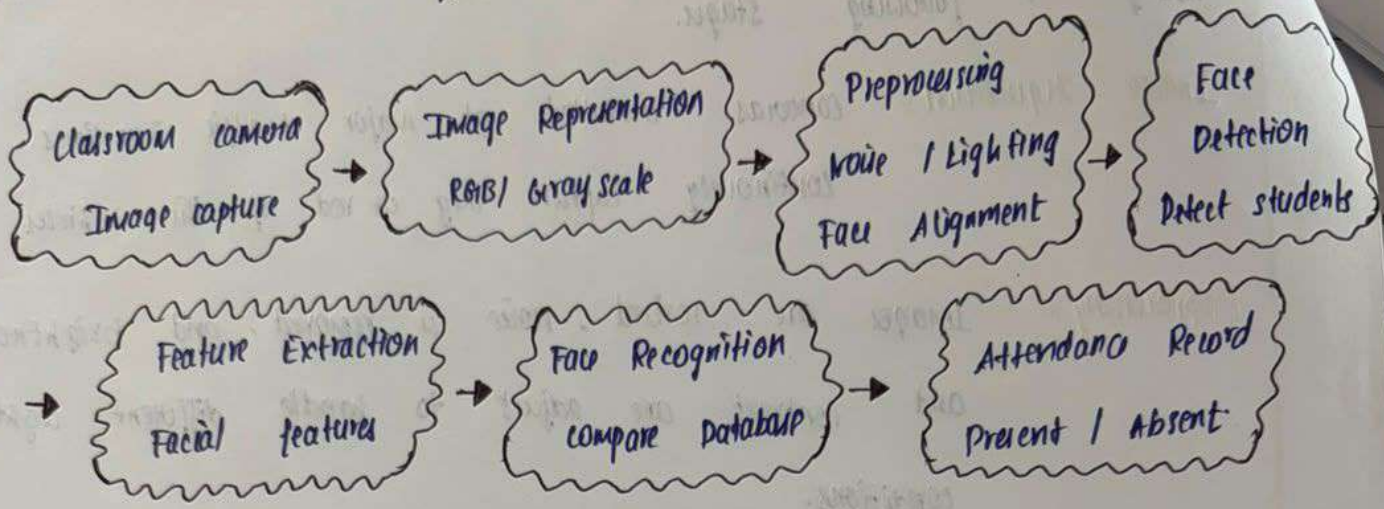
classification :- Detected objects are classified as cars, buses, motor cycles, or pedestrians.

Result Analysis :- The system can count vehicles, estimate traffic density, and identify congested areas.

Pipeline :- Image Acquisition → Preprocessing → segmentation → Feature extraction → object detection → classification → Traffic analysis.

2). Automated Attendance system.

FACE RECOGNITION PIPELINE



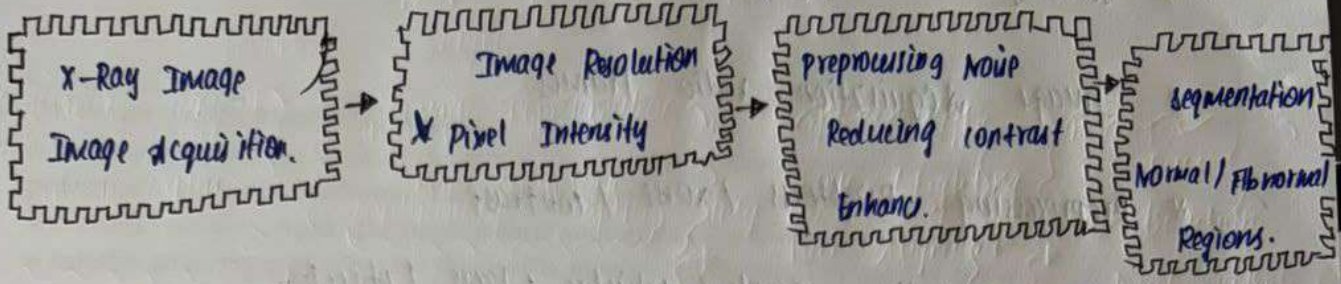
Explanation :

* Classroom captures images by camera containing multiple students.
* Images are represented using RGB / Grayscale Pixel values.
* Preprocessing improves brightness, removes noise, and faces.
* Face detection identifies individual faces.
* Important facial features are extracted from each detected face.
* If a student is recognized, attendance is automatically recorded.

Result : Automated attendance reduces manual work and provides faster attendance recordings.

Q3: Medical Image Analysis system.

X-Ray Analysis pipeline.



Resolution: Higher resolution provides more detailed information about tissues and abnormalities.

Intensity: Pixel intensity represents different brightness levels in x-rays

Preprocessing: Noise reduction, and contrast enhancement improve image quality.

Segmentation: Abnormal regions are separated from normal regions.

Feature extraction: Shape, texture, edges, and intensity patterns are extracted.

classification: The system classifies the region as normal or Potentially abnormal.

Result: computer vision assists doctors by highlighting suspicious regions and improving analysis efficiency.

Q4: Security surveillance system.

* Security cameras Entrance / wall / parking Areas
* Image Acquisition redio Frames
* Preprocessing Brightness / noise / contrast
* Object Detection People / vehicles / Bags / objects.
* Feature extraction shape / colour / Texture / size.
* classification Person / vehicles / other Objects.
* Object Tracking Movement analysis
* security Alert / monitoring.

Q5: smart Agriculture Monitoring.

* Segmentation plant / soil / weed / Disease
* Feature Extraction / colour / Texture / shape / size.
* classification / Healthy / Disease / weed / Damaged.
* crop Health Map * Decision Making.

challenges:
* Shadows from Plants and objects
* Different plant size
* Overlapping leaves.
* Changing sunlight conditions.

Result: The system enables large-scale crop monitoring and early identification of disease and damaged plants.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, in-depth4, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature:	Signature:	Signature:
Date: 07/10/26	Date:	Date: